

Integrated Climate-intelligent Salt Production Ecosystem: A Multi-modal AI Framework For Sustainable Yield Optimization In Tropical Coastal Environments

Project ID: 25-26J-431

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IT22308016 | SIRIMANNA R.D.I.B

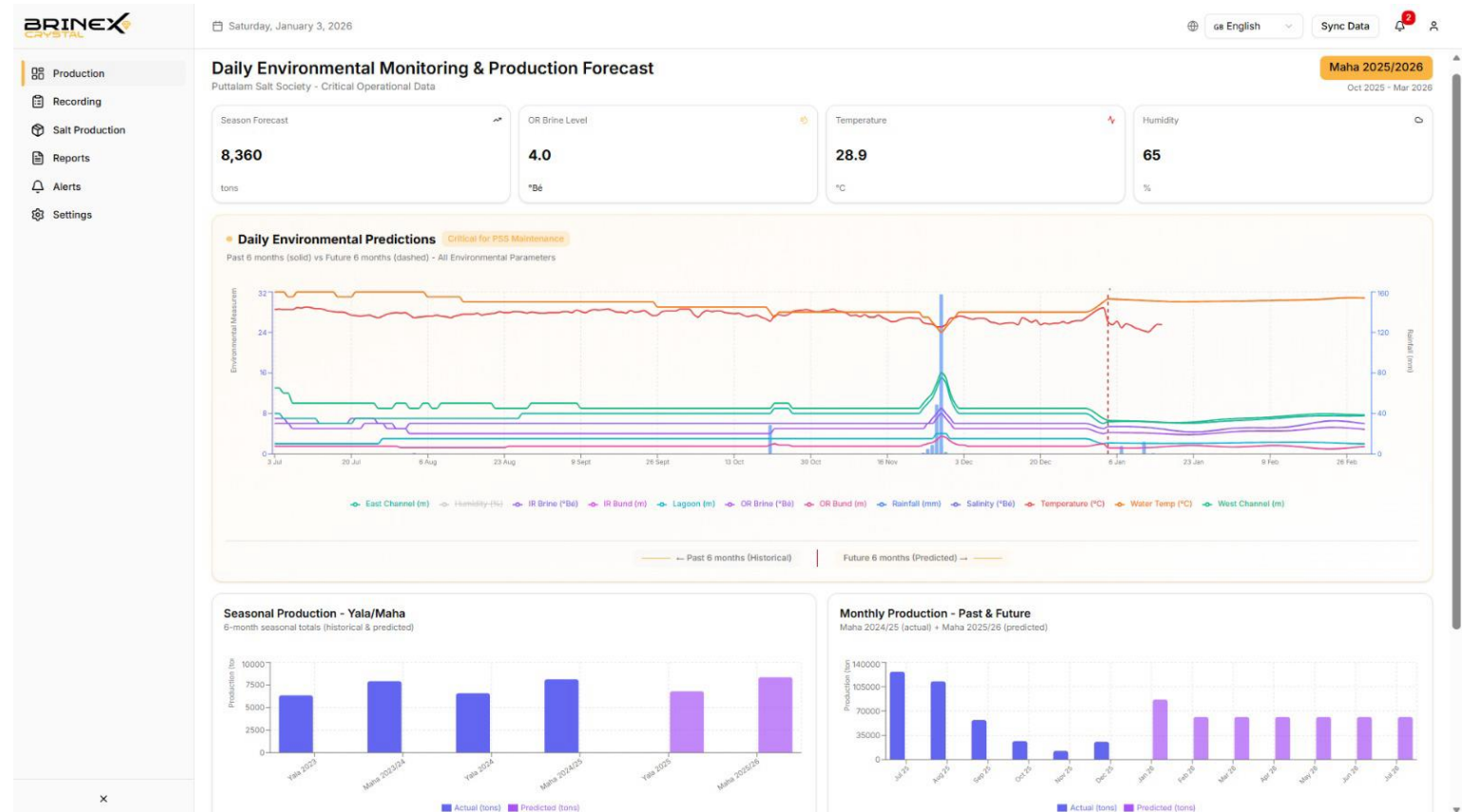
Crystallization Forecasting

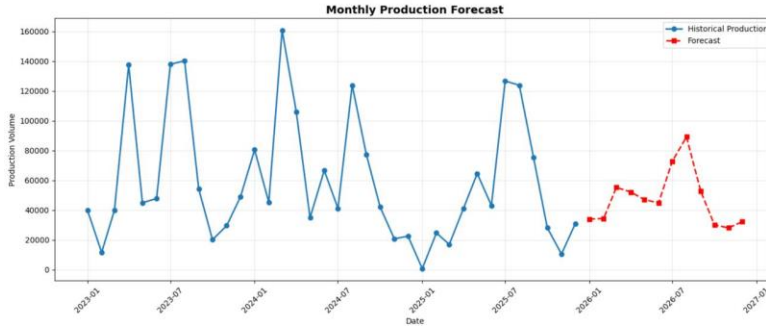
Software Engineering

Crystallization Forecasting

How the Solution will address the Sub-Problem / Prototype

- Sub Problem:
 - Unreliable Physical record system.
 - Harvesting decisions rely on basic weather alerts.
 - Cannot predict the best harvest time with high production.
- Solution:
 - Predicts harvest-ready salt production for the next 6 months
 - Use a hybrid ARIMA+LSTM model instead of manual estimates.
 - Automated Localized analytical Dashboard for PSS and Landowners.





Crystallization Forecasting

User Requirements addressed by the solution

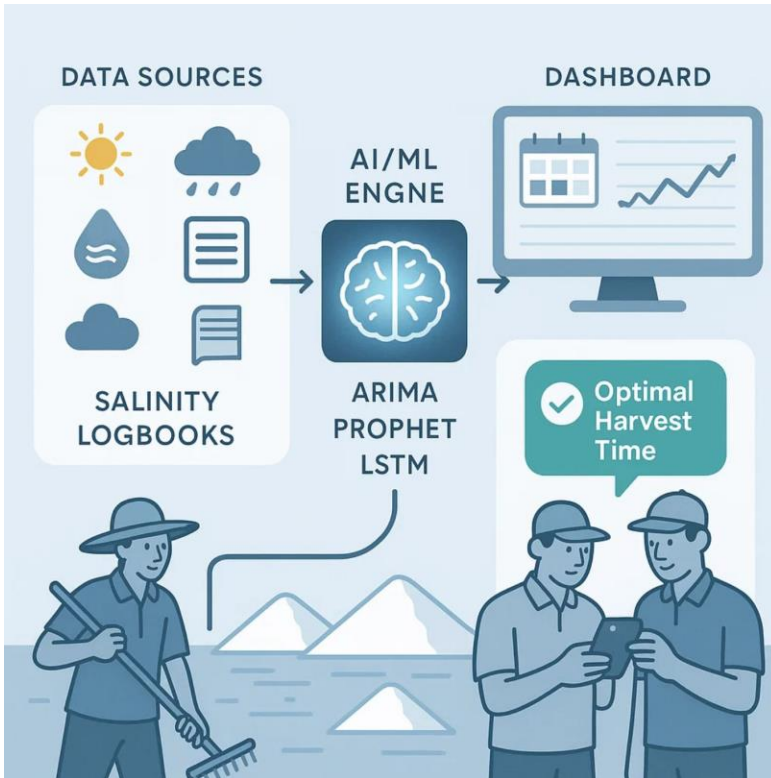
- Supports labour and pond planning
- Monthly/Seasonal production forecasts under changing rainfall and climate.
- Simple Sinhala/Tamil/English interface.

Design Excellence/Contribution

- Hybrid ARIMA+LSTM model
- tested standalone Prophet and XGBoost regressors
- Localization in the frontend
- Act as a Central service (used by valorization and planning)

User Feedback on Prototype

- Puttalam Salt Society confirmed they mostly produce standard table salt, so the main need is total production forecasting, not per crystal type.
- Users preferred having Tamil and Sinhala language support.





IT22893734 | ANSAR T. D

Intelligent Planning Engine

Software Engineering

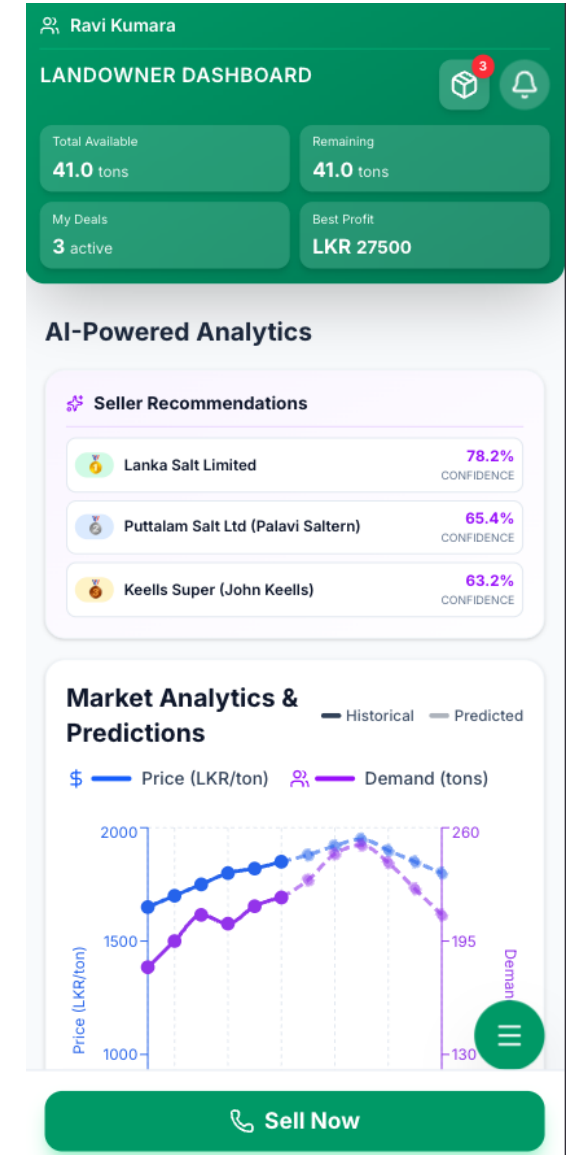
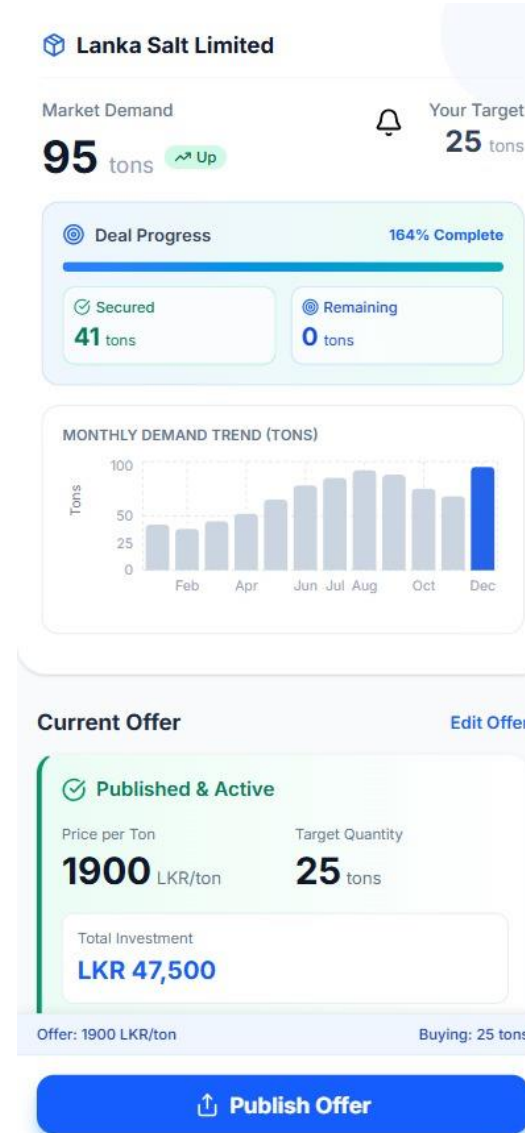
Intelligent Planning Engine

How the Solution will address the Sub-Problem / Prototype

- Sub Problem:
 - Landowners manually call and contact each seller
 - No real-time market visibility
 - Decisions based on incomplete data like who to sell to.
 - Cannot adapt to market demand changes
- Solution:
 - Seller Dashboard - Real-time offer publishing
 - AI Seller Matching - CatBoost recommendations
 - Price Forecasting - Market demand predictions
 - Smart Planning from Mobile

User Requirements addressed by the solution

- Eliminates manual seller calls (saves time and effort)
- Real-time offer visibility (central dashboard view)
- Faster, data-driven deal-making (instant recommendations)
- Avoids missing better offers (automated matching system)



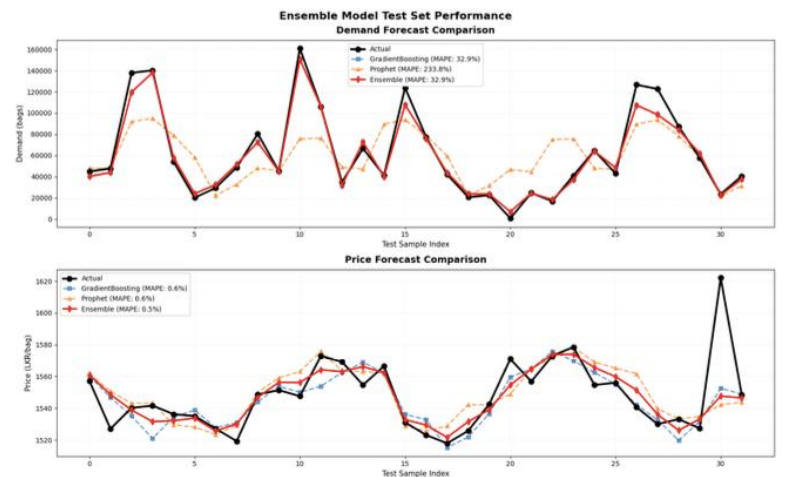
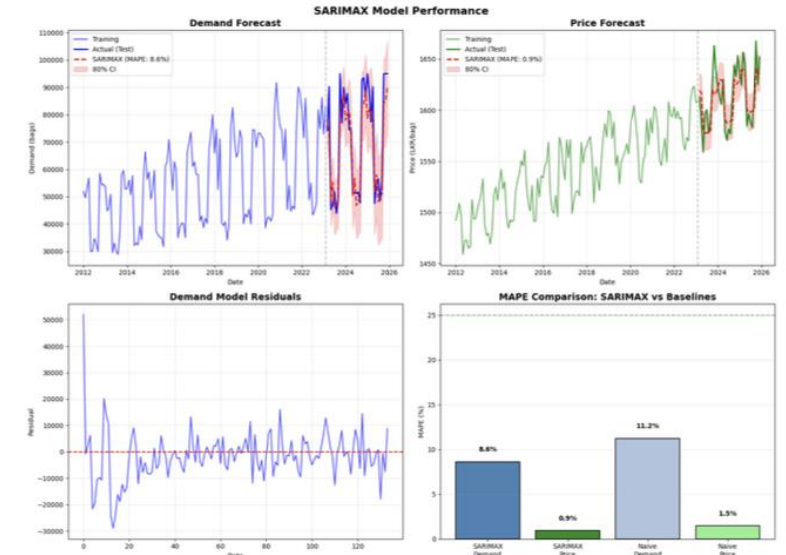
Intelligent Planning Engine

Design Excellence/Contribution

- Seller Recommendation Model
 - RL Model Challenge: 50% dataset coverage
 - CatBoost Solution: gradient boosting tree model
- Price/Demand Prediction
 - Current Model: SARIMAX - statistical time-series model
 - Tested: Gradient Boosting + Prophet
- Contribution: Replace manual calls with automated, data-driven seller and harvesting decisions.

User Feedback on Prototype

- The landowners mentioned the hardship of getting the best offers from sellers. A unified platform would be very useful and easy for their decision-making.
- User requested Language support.
- Requested to simplify the dashboard since neither the landowners nor the sellers are familiar with technology.





IT22346322 | ARSHAQ M.J.M

Computer Vision Quality Control System

Software Engineering

Computer Vision Quality Control System

- **Problem I:**

Slow & Inconsistent Manual Inspection - Human visual inspection is time-consuming, subjective, and prone to fatigue errors.

- **Solution:**

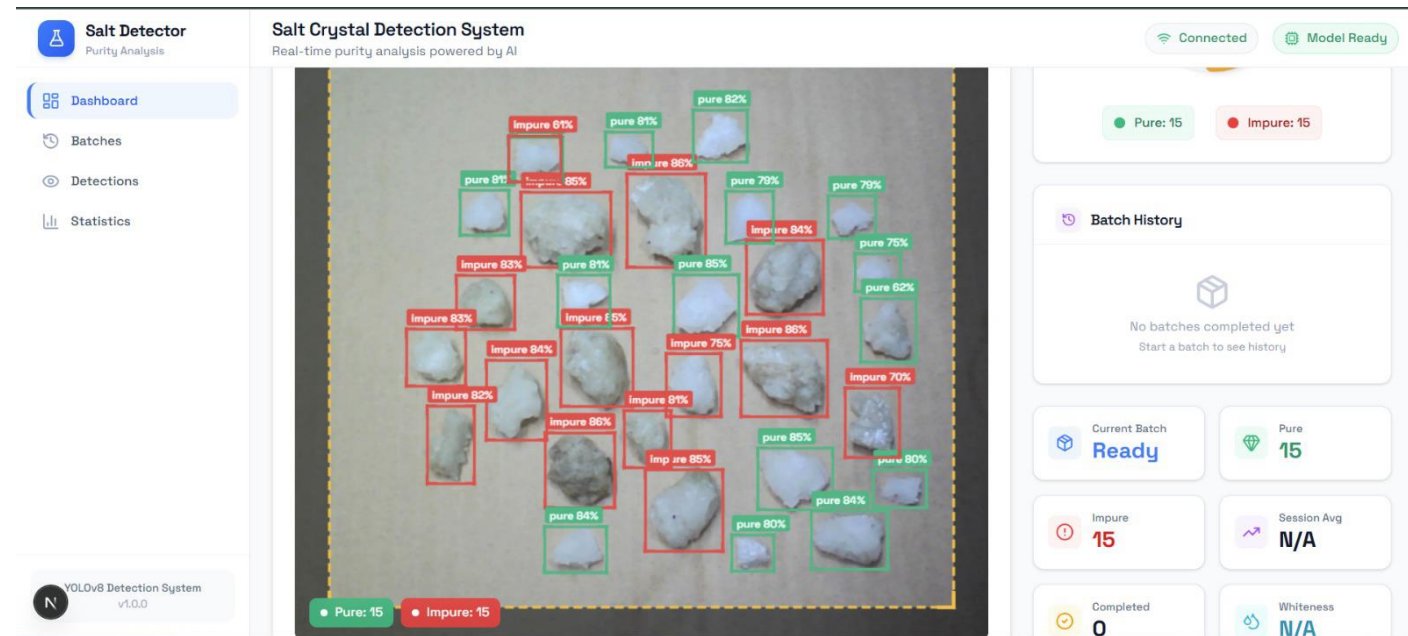
YOLOv8 AI model provides automated real-time detection in 50-100ms per frame with consistent accuracy, enabling 24/7 operation.

- **Problem II:**

No Real-Time Monitoring or Traceability - Quality issues are only discovered after batch completion, causing waste and delayed fixes. No systematic quality records exist.

- **Solution:**

Live dashboard with real-time purity tracking, batch logging with timestamps, and trend analysis for early detection of quality issues.



Computer Vision Quality Control System

Key Solutions Addressing User Requirements

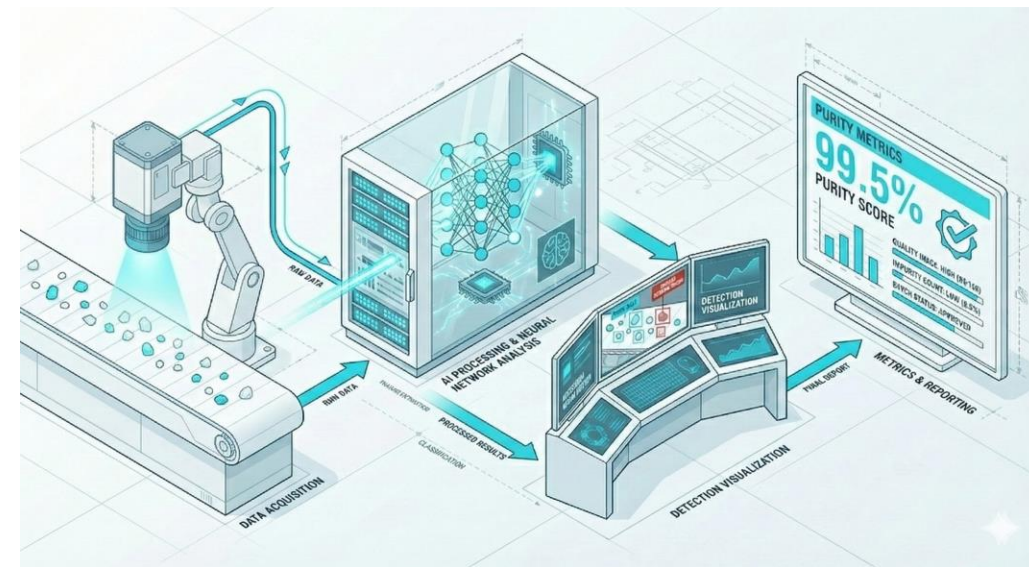
1. Real-Time Detection & Visual Feedback: Live camera feed with color-coded bounding boxes, real-time purity counters.
2. Batch & Session Tracking: Automated and manual batch creation with session management, historical comparison, and filterable detection history.
3. Analytics Dashboard: Comprehensive statistics with 30-day trend charts, and hourly distribution analysis.

Design Excellence and Contribution

- Technical Innovation: Browser-to-visualization pipeline with <200ms latency, dual-metric quality assessment (YOLOv8 + HSV analysis), and ROI-based smart filtering for accurate detection.
- Engineering Excellence: Modern stack (Next.js 14, NestJS, PostgreSQL, ONNX Runtime) with WebSocket real-time architecture and scalable indexed database for high-volume logging and analytics.

User Feedback on Prototypes

- Positive: Users praised detection accuracy (matching human inspectors), clear real-time visualization, and automated batch tracking that reduced manual record-keeping.
- Improvements: Future enhancements include camera auto-calibration, GPU acceleration for high-throughput, and PDF/Excel export for compliance.





IT22354310 | PERUMBULI P.G.R.M.D

Federated Learning for Waste Valorization

Software Engineering

Federated Learning for Waste Valorization

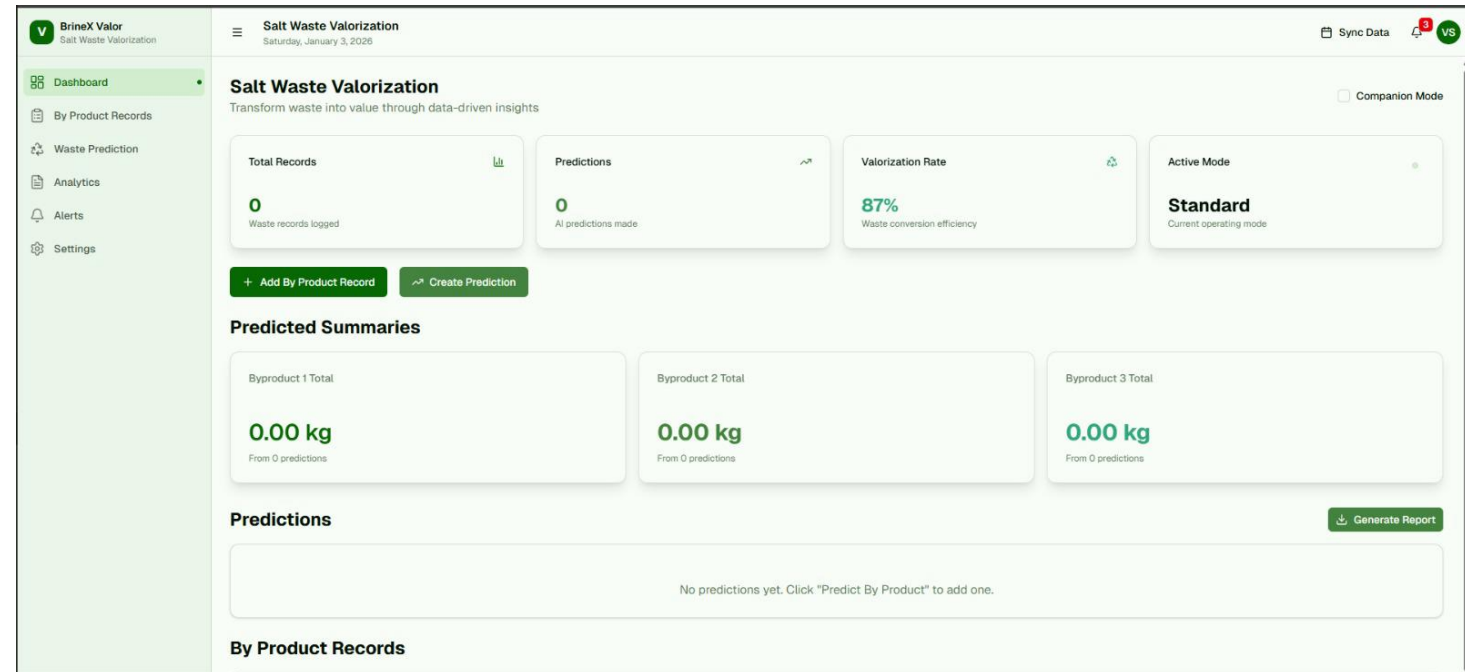
How the Solution will address the Sub-Problem / Prototype

- Sub Problem:
 - Salt production generates **20–30% waste** (gypsum, magnesium, organic matter)
 - Facilities **cannot share data** due to privacy & competition
 - Lack of accurate **waste composition prediction**
 - No collaborative optimization across multiple sites

- Solution:

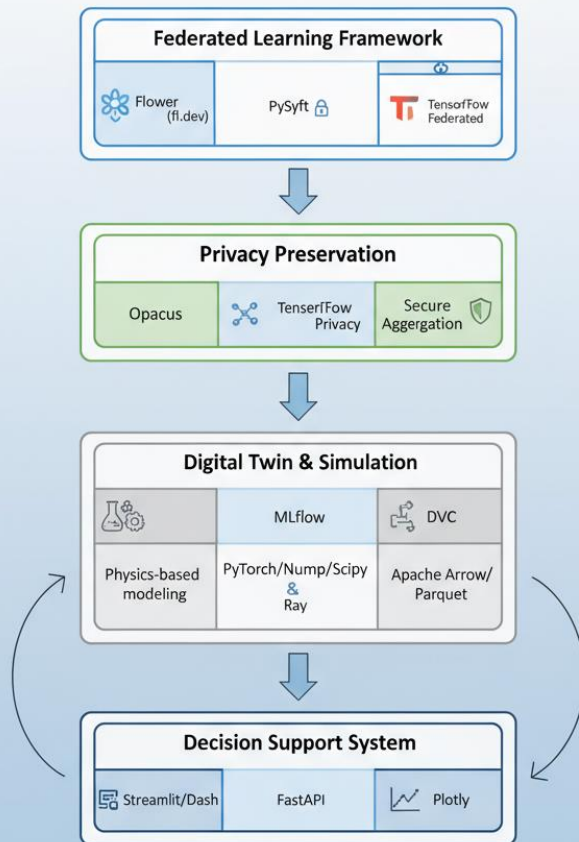
Federated Learning–based Digital Twin System

- Privacy-Preserving Learning
- Accurate Waste Prediction
- Production Optimization
- Decision Support



Federated Learning for Waste Valorization

Technology Stack for Federated Waste Valorization



User Requirements Addressed

- Waste prediction
- advance planning

Design Excellence & Novel Contributions

- First application of Federated Learning for salt waste management
- Integration of Digital Twins with Federated Learning
- Multi-model strategy
- Inverse Optimization
 - Finds optimal production conditions from waste targets
- Privacy-by-Design (Federated Learning)
 - 0 bytes of raw data shared

User Feedback on Prototypes

- Users valued the privacy-preserving approach and improved prediction accuracy as more participants joined, making it well-suited for cooperative societies like Puttalam Salt Producers.
- Future Enhancements: Simplified waste value visualization

Integrated Climate-Intelligent Salt Production Ecosystem

Commercialization & Sustainability

- Customer Persona & Target Market
 - Puttalam Salt Society
 - Landowners
 - Salt Distributors / Sellers
- Market Size
 - Initial: Puttalam region (the largest salt production hub in Sri Lanka)
 - Scale: Across all salt-producing regions in Sri Lanka (Hambanthota) by implementing tenants.
 - Export: Tropical regions in South Asia with similar salt production models
- What is Unique in Your Solution?
 - First AI-powered digital platform specifically designed for traditional salt production.
- How will the cost be recovered?
 - We'll use a SaaS subscription model with Tiers for the services.





Thank you.

It's time for Q&A